

This is sample data provided for CALIBER purposes only		ONE POINT LESSON (OPL)		OPL No: OPL-FA-8901-02 Sheet 1 of 1
OPL Title	PSV-8901 Testing & Car-Seal Management	Discipline	Mechanical	
Equipment	FA-8901 - REFLUX ACCUMULATOR DRUM	Area / Unit	8900 - HEXANE COLUMN OVERHEAD SYSTEM	
Related Interlock	SEQ-8901 (SIL 1)	P&ID; Ref	TJC-LLD-PID-8901	
Classification:	☒ Basic Knowledge	☐ Improvement	☐ Trouble Case	Aspect: P-Q-C-D-S-M-E

1. PURPOSE / OBJECTIVE

This One Point Lesson explains **psv-8901 testing & car-seal management** for FA-8901 (REFLUX ACCUMULATOR DRUM) in the HEXANE COLUMN OVERHEAD SYSTEM. The relief valve is the drum's last line of overpressure defence and must be proven and never isolated in service. The lesson gives operators and technicians the key knowledge and the correct step-by-step method so the task is done safely, correctly and consistently every time.

2. SAFETY PRECAUTIONS

Hazard note: this equipment is **HIGH CRITICAL** handling **Hexane / water** at up to 10 barg / FV / 120 degC. Observe the following before and during the task:

- Apply LOTO (Lock-Out Tag-Out) and prove zero energy before starting work.
- Isolate, depressurise, drain and purge the equipment; obtain a valid work permit.
- Wear the correct PPE: coverall, safety shoes, gloves, goggles and hearing protection.
- Beware of stored mechanical energy (springs, rotating inertia) and pinch points.
- Use certified lifting gear and correct manual-handling technique for heavy parts.

3. TOOLS & MATERIALS REQUIRED

- Standard mechanical hand tools & calibrated torque wrench
- LOTO padlocks and permit
- Laser/dial alignment kit (as applicable)
- Feeler gauges, dial indicator, seal/bearing pullers
- Correct spare parts and gaskets per the BOM
- Lifting gear and slings (certified)

4. DETAILED PROCEDURE / STEPS

Step	Action	Check / Acceptance
1	Confirm PSV-8901 is set to 10 barg	Isolation & LOTO verified
2	Carry out the off-line pop test per the statutory schedule	Parts match BOM / spec
3	Keep the PSV isolation car-sealed open in service	Torque / clearance to spec
4	Tag and file the test certificate	No leak / smooth operation
5	Never block-in a live PSV without an authorised spare	Record in equipment history

5. COMMON PROBLEMS & TROUBLESHOOTING

Symptom	Likely Cause	Action
Statutory relief valve test	Set pressure drift found	Overhauled PSV-8901, lapped seat, re-set 10 barg, certified
Boot drain valve N4 passing / leaking	Seat eroded by water/solids	Reseated / replaced boot drain valve HV-8904
Scheduled corrosion monitoring of boot	Wall thickness within corrosion allowance	UT survey of boot and shell bottom, logged in RBI file

6. KEY LEARNING POINTS

- The relief valve is the drum's last line of overpressure defence and must be proven and never isolated in service.
- Respect the interlock SEQ-8901: e.g. LSHH-8901 trips at > 85 % - never defeat it without a permit.
- Always cross-check the Datasheet limits and the P&ID; TJC-LLD-PID-8901 before working on FA-8901.
- Record the work in the equipment history so the Knowledge Hub stays the single source of truth.

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